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Robust Methods for Detection of Shifts of the Innovation Variance of a Time Series

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Based on Pitman asymptotic relative efficiency and Monte Carlo studies of power functions, Shorack [29] concluded in the independent and identically distributed (i.i.d.) case that the most satisfactory tests for scale changes were the Box-Andersen test and the jackknife. Analogs of these two procedures are shown to have identical asymptotic properties when the observations follow a correctly identified autoregressive process. A Monte Carlo study indicates that the performance of these robust procedures is roughly the same as in the i.i.d. case. The difference between the two robust procedures seems slight. The Box-Andersen procedure does a better job of maintaining the stated significance level but it lags slightly behind the jackknife in power.

KEY WORDS

Autoregressive process
Variance shift
Robust test
Jackknife
Asymptotic distribution
Monte Carlo

1. INTRODUCTION

The problem of detecting shifts of parameters of a time series has received considerable attention recently. (See Bagshaw and Johnson [5] for a summary of recent work.) This paper studies robust methods for detecting a shift in variance at a specified point of a time series.

Detecting shifts of variance can be useful in finance where the variance is used as a measure of risk [15]. In particular it is useful in estimating the current variance of a stock price series, which is used in Markowitz's optimal portfolio selection procedure [21]. In physical sciences it could be used to determine if the accuracy of a record has changed at a specified time. As an example, in Section 6 the precision of early measurements of Wolfer's sunspot numbers are compared with more recent measurements.

The suspected shift point divides the observations into two regimes. If the observations are assumed to

be mutually independent, the usual two sample tests can be applied to the two regimes. Under the additional assumption that the data are normally distributed, the likelihood ratio test of equality of variances is the F test. It is well known that the distribution of this test statistic depends quite heavily on the normality assumption. Indeed, Box [6] suggested that this procedure could be used as a test of normality rather than equality of variances.

Box and Tiao [9], Section 3.2) distinguish between criterion robustness and inference robustness. If a criterion is little affected by non-normality, this type of sensitivity is referred to as criterion robustness with respect to non-normality. If the data are not normally distributed, the F test is no longer the likelihood ratio test for equality of variances. In this case the test should be made using some other procedure. If the inference (acceptance or rejection of the hypothesis using the likelihood ratio test) does not change with small deviations from normality, there is inference robustness with respect to deviations from non-normality.

Most work on robustness has dealt with criterion robustness. If a robust criterion is employed, it allows the data analyst the luxury of checking one less assumption. This paper is concerned with criterion robustness of tests of equality of variances with respect to deviations from normality.

In the independent and identically distributed (i.i.d.) case, there have been a large number of papers proposing and comparing robust criteria for testing equality of variances. Shorack [29] compared several

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robust criteria on the basis of Pitman asymptotic relative efficiency (a.r.e.) and Monte Carlo studies of power functions. He concluded that an extension of the Box-Andersen permutation test [7] and the jackknife procedure (Miller [24]) were the most satisfactory.

It is quite possible that some of the ideas contained in the Princeton Robustness Study [3] could be applied to yield robust criteria for testing equality of variances. A Monte Carlo study by Brown and Forsythe [10] showed that a trimmed version of Levene's test [19] was quite robust.

Much less is known about robust criteria when the observations are correlated. When an i.i.d. sample is available on a p dimensional random variable, tests for the equality of the p variances have been studied by Han [12] and by Levy [20]. For univariate time series, previous work on detecting shifts of variance was given by Herbst [13] for moving average models and by Wichern, Miller, and Hsu [32] for autoregressive models. The latter two studies restricted attention to Gaussian processes.

The model studied in this paper allows a shift of the mean, variance, and the autoregressive parameters at the specified time point. This paper gives robust procedures for detecting a variance shift. In particular, we show that analogs of the Box-Andersen procedure and the jackknife have identical asymptotic properties to the i.i.d. case when the observations follow an autoregressive process. This implies that the Pitman a.r.e. is the same as in the i.i.d. case.

It is assumed throughout this paper that the order of the autoregressive process is known and is the same on both regimes. By reading through the proofs it is clear that the asymptotic results continue to hold when the order of the process also shifts. For notational simplicity we assume that the order is the same on both regimes. In applications the order of the process will also be an estimate. This paper does not consider the effect that the estimation of the order of the process has on the test for shift in variances.

The theory developed here for testing for a single variance shift can be extended in a straightforward fashion to the case of multiple shift of parameters. Extensions of the q sample procedures studied by Layard [18] can be used to test for the homogeneity of the q variances on the time intervals formed by the time points $T_1 < T_2 < \dots < T_q$.

2. MODEL

We assume that after an appropriate transformation of the data we have a univariate time series, y_t , $1 - p \leq i \leq T$, which is a sample from an autoregressive process of known order, p . The mathematical

model can be stated

$$\left. \begin{aligned} \varphi_1(B)y_t &= \nu_1 + a_t \quad t \leq T_1 \\ \varphi_2(B)y_t &= \nu_2 + a_t \quad t > T_1 \end{aligned} \right\} \quad (2.1)$$

where $\varphi_i(B) = 1 - \sum_{j=1}^p \varphi_{ij} B^j$ for $i = 1, 2$ and B is the backshift operator which satisfies $By_t = y_{t-1}$. If there are changes in the parameters of (2.1), the time series y_t is clearly nonstationary. We will assume that the autoregressive parameters correspond to stationary processes in the sense that all roots of $\varphi_i(B)$ be outside the unit circle in the complex plane.

We assume that the standardized innovations a_t are an independent sample from a distribution with cumulative distribution function (c.d.f.) G . That is, a_t has c.d.f. $G(x/\sigma_1)$ for $t \leq T_1$ and c.d.f. $G(x/\sigma_2)$ for $t > T_1$. We will not specify G completely but will assume that it has the following moments

$$\int x dG(x) = 0, \quad \int x^2 dG(x) = 1, \quad \text{and} \quad \int x^4 dG(x) = \gamma_2 + 3$$

where the kurtosis $\gamma_2 < \infty$.

The hypothesis of interest is to test

$$H: \Delta^2 = 1 \quad \text{vs.} \quad A: \Delta^2 \neq 1$$

where $\Delta^2 = \sigma_2^2/\sigma_1^2$.

In Section 3 we derive the likelihood ratio test for H under the additional assumption that G is the normal c.d.f. In Section 4 it is shown that the likelihood ratio test does not possess criterion robustness with respect to deviations from normality. The asymptotic distribution of the jackknife and the Box-Andersen procedure are stated here. The results of a small Monte Carlo study are presented in Section 5. In Section 6 the techniques are applied to the Wolfer sunspot series.

3. LIKELIHOOD RATIO PROCEDURE

In this section we make the further assumption that G is the standard normal c.d.f. Even in the stationary case, the likelihood function for autoregressive processes is somewhat complicated. If the first p "starting values" of the series are assumed given, the likelihood is more tractable. In this section we give the likelihood ratio test of H assuming that the starting values are given.

If $\| \cdot \|$ denotes Euclidean distance and we define "regression" parameters β_i by $\beta_i' = (\varphi_{i1}, \varphi_{i2}, \dots, \varphi_{ip}, \nu_i)$ for $i = 1, 2$, the conditional density of the vector $y = (y_1, \dots, y_T)$ given the observations $w = (y_0, \dots, y_{1-p})$ and the parameters $\theta = (\beta_1', \beta_2', \sigma_1^2, \sigma_2^2)$ is

$$f(y|w, \theta) = \exp \left\{ - \sum_{i=1}^2 \frac{\|y_{(i)} - X_i \beta_i\|^2}{2\sigma_i^2} \right\} / \left((2\pi\sigma_1^2)^{-T_1/2} (2\pi\sigma_2^2)^{-(T-T_1)/2} \right)$$

where $y'_{(1)} = (y_1, y_2, \dots, y_{T_1})$, $y'_{(2)} = (y_{T_1+1}, \dots, y_T)$, $X'_1 = (x_0, x_1, \dots, x_{T_1-1})$, $X'_2 = (x_{T_1}, x_{T_1+1}, \dots, x_{T-1})$, $x'_i = (y_i, y_{i-1}, \dots, y_{i-p+1}, 1)$, and $T_2 = T - T_1$. If τ is the likelihood ratio statistic for H , it is easy to show

$$-2 \ln \tau = \sum_{i=1}^2 T_i \ln \hat{\sigma}_i^2 / \hat{\sigma}^2$$

where

$$\hat{\sigma}_j^2 = \|y_{(j)} - X_j \hat{\beta}_j\|^2 / T_j,$$

$$\hat{\beta}_j = (X_j' X_j)^{-1} X_j' y_{(j)},$$

$$\hat{\sigma}^2 = \sum_{j=1}^2 \|y_{(j)} - X_j \hat{\beta}_j\|^2 / T = c \hat{\sigma}_1^2 + (1 - c) \hat{\sigma}_2^2$$

where $c = T_1 / (T_1 + T_2)$. The likelihood ratio test is seen to depend only on $F = \hat{\sigma}_2^2 / \hat{\sigma}_1^2$.

4. LARGE SAMPLE DISTRIBUTIONS

The effect of non-normality on the F statistic is asymptotically the same as in the i.i.d. case. In the Appendix (Lemma 3) we show that the approximate large sample distribution of F under the model of Section 2 is

$$F \overset{app}{\sim} N(\Delta^2, \Delta^4(2 + \gamma_2)(T_1 + T_2)/T_1 T_2).$$

If a consistent estimate is obtained for γ_2 , an asymptotically robust procedure can be obtained using this approximate distribution. Monte Carlo work in the i.i.d. case suggests that the Box-Andersen procedure and the jackknife should have better properties [25, 29].

An extension of the Box-Andersen procedure is obtained by approximating the distribution of F by the F_{dv_2, dv_1} distribution where

$$v_i = T_i - (p + 1) \quad i = 1, 2,$$

$$d = (1 + (b_2 - 3)/2)^{-1},$$

$$b_2 = (T + 2) \sum_1^T \hat{a}_i^4 / \left(\sum_1^T \hat{a}_i^2 \right)^2,$$

and the i th residual $\hat{a}_i$ is obtained from (2.1) by replacing the parameters with their estimates.

Shorack showed that in the i.i.d. case the Box-Andersen test and the F test have Pitman a.r.e. 1 with respect to each other. Lemma 4 in the Appendix shows that this is also true for autoregressive processes.

The other method advocated by Shorack in the i.i.d. case was the jackknife, which seems more powerful than other data splitting techniques such as the one proposed by Box [6]. Suppose that the integers in the j th group ($j = 1$ and 2) are partitioned into k_j groups each of size m where $T_j = k_j m$. Label these $k_1 + k_2$ sets of integers as $s_{i(j)}$ for $i = 1, \dots, k_j$ and $j = 1, 2$.

Two methods have been proposed for two sample jackknife problems. The method followed here is to jackknife each "sample" separately and combine the results. An alternative method is to successively delete observations from the first sample leaving the second sample intact then reverse the roles of the two samples. For estimation of variance in the i.i.d. case, Miller [25] used the former method while Arvesen's [4] Theorem 16 could be used to substantiate the latter method. Miller [26] has pointed out that the question of superiority between these two procedures has not been settled.

We assume that g is a function with a bounded second derivative in a neighborhood of σ_i^2 . The jackknife is based on pseudovalues

$$\hat{\theta}_{i(j)} = k_j g(\hat{\sigma}_j^2) - (k_j - 1) g(v_{i(j)})$$

$$i = 1, \dots, k_j \quad j = 1, 2$$

where

$$v_{i(j)} = \sum_{1 \neq t} \sum_{t \in s_{1(j)}} \hat{a}_t^2 / m(k_j - 1).$$

Although we are jackknifing a mean, the results of Arvesen on jackknifing U -statistics do not apply since the summands (the residuals) are dependent. The theorem below shows that the residuals are close enough to independent random variables for the jackknife theorem to hold. The jackknife estimate of $g(\sigma_j^2)$ is

$$\hat{\theta}_{(j)} = \sum_{i=1}^{k_j} \hat{\theta}_{i(j)} / k_j$$

and the residual sum of squares is

$$w_j = \sum_{i=1}^{k_j} (\hat{\theta}_{i(j)} - \hat{\theta}_{(j)})^2 \quad j = 1, 2.$$

The jackknife theorem is stated below using the variance stabilizing log transformation which has proven to be useful for inference on variances.

Theorem: Assume $g(u) = \log u$ then

(i) If $m = 1$, $K_j = T_j$, and $\min\{T_1, T_2\} \rightarrow \infty$

$$\frac{(\hat{\theta}_{(2)} - \hat{\theta}_{(1)} - \log \Delta^2)}{\left(\sum_1^2 w_i / T_i (T_i - 1) \right)^{1/2}} \xrightarrow{L} N(0, 1).$$

(ii) If k_1 and k_2 are fixed and $m \rightarrow \infty$

$$\left(\frac{k_1 k_2}{k_1 + k_2} \right)^{1/2} \frac{(\hat{\theta}_{(2)} - \hat{\theta}_{(1)} - \log \Delta^2)}{\left(\frac{w_1 + w_2}{k_1 + k_2 - 2} \right)^{1/2}} \xrightarrow{L} t_{k_1 + k_2 - 2}.$$

Proof: Since the proof is basically the same as Miller's original one (Miller [24], Theorems 1 and 2), it is only sketched in the Appendix for part (i).

Monte Carlo and relative efficiency results in the i.i.d. case suggest that information is lost by increas-

ing m . For long time series, it may prove convenient to pick $m > 1$. The theorem suggests if $m > 1$ a Student- t distribution should be used as an approximate distribution. In the i.i.d. case Hinkley [14] proposed alternative degrees of freedom for the t approximation based upon sampling moments.

5. MONTE CARLO STUDY

In order to study the small sample power properties of the tests, a small simulation was conducted. Time series of length 60 were generated from an $AR(1)$ model. If the time series contained parameter shifts, these occurred after 30 observations making the two regimes of equal length. To facilitate comparison with the Monte Carlo study of Miller [25], the values of Δ^2 considered were 1, 2, 4, and 6 and the error distributions considered were uniform, normal, and double exponential. Pseudo-normal random variables were generated using the probability mixing algorithm KR ; see Kinderman and Ramage [16].

Each series was generated 2000 times to estimate the probability of detecting a shift using significance level .05. The results of the simulation are presented in Table 1. Since there are 30 observations in each group, the power of the tests are naturally higher than Miller's study with 25 in each group. The general conclusions are the same, though. For these error distributions, the Box-Andersen procedure does the best job of maintaining the nominal significance level, but its power lags behind the jackknife with $m = 1$.

Of course, it is unreasonable to compare power functions of tests that have different significance lev-

els. It could be argued that the power of the Box-Andersen procedure is smaller than the jackknife, because the jackknife has a larger significance level. When the significance level is determined by simulation, it is difficult to determine if this is the case. In any event the difference between these two procedures seems slight, at least for the deviations from normality studied here.

The results reported in Table 1 are for an $AR(1)$ model with mean = 0 and $\varphi = .5$. For large samples the power of these tests should not depend on (β_1, β_2) . At least in the case $p = 1$, additional Monte Carlo work indicates that regimes of length 30 are sufficient to guarantee that this is the case.

6. ILLUSTRATION

The data used in this study are yearly averages of the Wolfer sunspot numbers from 1700–1960 (see Waldmeier [30], pp. 20–21). For the period 1749–1960 monthly means are available while for 1700–1748 only yearly means exist. Waldmeier posits [p. 8] that the ancient data are not as reliable as the present data. We test this hypothesis by determining if the noise variances of the time periods 1700–1748 and 1749–1960 are significantly different.

Attempts to model Wolfer's sunspot numbers go back to Yule [33] who fitted an $AR(2)$ to the series. Box and Jenkins [8] also used an $AR(2)$ for the period 1770–1869. Examination of the partial autocorrelation function suggests that lags higher than 2 may prove useful. The autocorrelation and partial autocorrelation function of the two regimes of the sunspot

TABLE 1—Monte Carlo power functions for tests on variances.

Δ^2	1	2	4	6
<u>Uniform Distribution</u>				
F	.005	.402	.992	1
Box-Andersen	.042	.706	.996	1
Jackknife $m=1$	.037	.756	.998	1
Jackknife $m=5$	.048	.696	.994	1
<u>Normal Distribution</u>				
F	.056	.429	.948	.996
Box-Andersen	.059	.410	.926	.992
Jackknife $m=1$	.068	.450	.942	.994
Jackknife $m=5$	.066	.426	.907	.986
<u>Double Exponential Distribution</u>				
F	.190	.456	.873	.966
Box-Andersen	.048	.228	.661	.854
Jackknife $m=1$	.074	.278	.702	.882
Jackknife $m=5$	.074	.257	.660	.840

data are given in Table 2. Autoregressive models containing lags higher than 2 have been considered by Whittle [31] and by Akaike [1].

In this paper the order of the autoregressive process on each of the two regimes was estimated using methods due to Akaike [1] and to Schwarz [28]. These criteria choose the order p of the autoregressive process by finding the minima of the following quantities,

$$\text{Akaike: } 2p + n \ln \hat{\sigma}_p^2$$

$$\text{Schwarz: } p \ln n + n \ln \hat{\sigma}_p^2,$$

where n is the length of series, p is the order of the process, and $\hat{\sigma}_p^2$ is the estimated variance. Schwarz points out that his criterion leads to more parsimonious models if $n \geq 8$.

The order of the process was estimated on each of these two regimes using these two criteria. The two methods are consistent in the sense that they pick the same orders for both regimes. This is somewhat surprising since the regimes differ considerably in length. The Schwarz method leads to Yule's original $AR(2)$ model on both regimes while the Akaike method leads to an $AR(9)$ on both regimes. The estimated coefficients of the models are given in Table 3.

For the $AR(9)$ model many of the coefficients are small. This suggests that a subset selection procedure may pick an $AR(9)$ model with some of the coefficients equal to zero [22, 23].

Using either of the models the residuals are clearly non-normal. The values of b_2 are 5.51 using the $AR(2)$ model and 5.91 using the $AR(9)$. These values suggest using an error distribution more similar to the double exponential than to the normal. Since normality is suspect, the p -value of the F test is unreliable.

The value of the F statistic and the jackknife ($m = 1$) are given in Table 4 for $p = 2$ and 9. This table also gives the two sided p -values of the F , Box-Andersen, and the jackknife tests. In both cases the two robust

methods give p -values greater than 0.05 while the p -value of the F -test is less than .05.

The asymptotic distribution of the F statistic can be used to show that the F -test rejects too often when there is positive kurtosis. Table 1 shows that this is the case for samples of size 30, at least for an $AR(1)$ model. Although the numbers may change slightly for different order autoregressive models, the same general conclusion would be obtained. Thus, it is reasonable that the F test rejects while the more trustworthy methods indicate that the variance has not shifted. Although Table 1 indicates for distributions with positive kurtosis that the jackknife rejects more often than the Box-Andersen procedure, the p -value with the jackknife is not necessarily smaller than the p -value for the Box-Andersen procedure. In fact, Table 4 shows for the $AR(9)$ model the p -value for the $B-A$ procedure is slightly smaller.

7. SUMMARY

This paper studies tests for a shift in variance at a specified point of an autoregressive process. The normal theory likelihood ratio procedure is derived. It is shown that this procedure does not have asymptotically the correct level of significance.

When $p = 0$ the problem of testing for a variance shift of an autoregressive process reduces to a two-sample test of equality of variances. In this case a number of procedures have been proposed which are robust with respect to deviations from normality. The Box-Andersen procedure and the jackknife, which have been advocated in the i.i.d. case, are extended to the case of the general autoregressive process. The asymptotic distributions of these procedures are shown to be the same as in the i.i.d. case. Thus, for the general autoregressive process, comparison of these procedures using Pitman a.r.e. is the same as in the i.i.d. case.

The finite sample characteristics of these tests are studied in a small simulation. For an $AR(1)$ model

TABLE 2—Autocorrelation function (ACF) and partial autocorrelation function (PACF) of sunspot series.

ACF		1	2	3	4	5	6	7	8	9	10
Regime 1	1-10	.81	.45	.05	-.27	-.44	-.42	-.23	.06	.36	.55
	11-16	.57	.42	.13	-.16	-.38	-.45				
Regime 2	1-10	.82	.44	.04	-.26	-.38	-.32	-.12	.17	.42	.56
	11-20	.53	.35	.10	-.12	-.25	-.28	-.21	-.09	.05	.17
PACF											
Regime 1	1-10	.81	-.61	-.21	-.02	-.01	.11	.18	.19	.20	-.05
	11-16	0	-.03	-.18	.06	0	-.05				
Regime 2	1-10	.82	-.67	-.08	.05	0	.16	.14	.22	.10	.01
	11-20	.04	-.05	.05	.03	-.05	-.08	-.08	-.09	.02	-.02

with 30 observations in each regime, the operating characteristics of the F , the Box-Andersen, and the jackknife procedure are studied for the uniform, normal, and double exponential error distributions. As expected, the F test is terribly nonrobust. The performance of the Box-Andersen procedure and the jackknife are roughly the same. The Box-Andersen procedure does a better job of maintaining the correct level of significance but lags slightly behind the jackknife ($m = 1$) in power.

These procedures are applied to the Wolfer sunspot series to determine if the early observations have the same accuracy as the more modern ones. The series is broken into two regimes where the division point 1749 is the first year that monthly means are reported. The order of the autoregressive process was selected using Akaike's criterion and Schwarz's method. The error distribution has kurtosis of greater than 2.5 in both cases. Thus, the p -value associated with the F test is quite suspect. In both cases the F test rejects at level .05 while the more robust procedures do not.

8. APPENDIX

Consider the model

$$\varphi(B)y_t = \nu + a_t \quad (8.1)$$

where $\varphi(B) = 1 - \sum \varphi_i B^i$ has all its roots outside the unit circle, a_t is an i.i.d. sequence with mean 0 and variance σ^2 ,

$$\beta' = (\varphi_1, \dots, \varphi_p, \nu), \quad X' = (x_0 \ x_1 \ \dots \ x_{T-1}),$$

$$x_t' = (y_t, y_{t-1}, \dots, y_{t-p+1}, 1),$$

$y' = (y_1, \dots, y_T)$, and $\hat{\beta}' = (X'X)^{-1}X'y$. The following conditions on (8.1) will be useful:

C_1 : (8.1) holds for all t ;

C_2 : (8.1) holds for $1 \leq t \leq T$ and x_0 is fixed;

C_3 : (8.1) holds for $1 \leq t \leq T$, x_0 is stochastic and satisfies $Ex_0'x_0 < \infty$.

Lemma 1: Under any of conditions $C_1 - C_3$ one has as $T \rightarrow \infty$ that

$$T^{1/2}(\hat{\beta} - \beta) \xrightarrow{L} N(0, \sigma^2 \Sigma^{-1}) \text{ and}$$

$$X'X/T \xrightarrow{P} \Sigma$$

where

$$\Sigma = \begin{pmatrix} F + \mu^2 1_p 1_p' & \mu 1_p \\ \mu 1_p' & 1 \end{pmatrix},$$

$\mu = \nu/\varphi(1)$, $F = (f_{ij}) = (\gamma_{|i-j|})$ where $\gamma_k = E(y_t - \mu)(y_{t-k} - \mu)$, and the expectation is taken with respect to a sequence y_t satisfying C_1 .

Proof: Anderson ([2], section 5.5) shows these convergences under C_1 and C_2 . By examination of his proof it is clear it holds for C_3 .

Corollary: Under the assumptions of Section 2 one has

$$T_i^{1/2}(\hat{\beta}_i - \beta_i) \xrightarrow{L} N(0, \sigma_i^2 \Sigma_i^{-1}) \text{ and}$$

$$X_i'X_i/T_i \xrightarrow{P} \Sigma_i,$$

where Σ_i is defined as in Lemma 1.

Proof: For $i = 1$ this follows from Lemma 1 using C_1 . For $i = 2$ the distribution of $\hat{\beta}_2$ depends on the first sample only through x_{T_1} . Since $Ex_{T_1}'x_{T_1} = 1 + p\mu_1^2 + T_1(F_1) < \infty$, one has the result for $i = 2$.

Let r_i denote the i th residual of the series obtained by replacing parameters by estimates in equation (2.1). Define vectors of errors $a_{(1)'} = (a_1, \dots, a_{T_1})$ and $a_{(2)'} = (a_{T_1+1}, \dots, a_T)$, vectors of residuals $r_{(1)'} = (r_1, \dots, r_{T_1})$ and $r_{(2)'} = (r_{T_1+1}, \dots, r_T)$, and sets of integers $s_{(1)} = (1, \dots, T_1)$ and $s_{(2)} = (T_1 + 1, \dots, T)$.

Lemma 2: The quantities $r_{(i)'}r_{(i)}$ and $r_{(i)'}a_{(i)}$ are bounded in probability for $i = 1, 2$.

Proof: By the corollary and Layard [17, Theorem 1.2] one has

$$r_{(i)'}r_{(i)} = (\hat{\beta}_i - \beta_i)'X_i'X_i(\hat{\beta}_i - \beta_i) \xrightarrow{L} \sigma_i^2 \chi_{p+1}^2$$

for $i = 1, 2$.

To show $r_{(i)'}a_{(i)}$ is bounded in probability consider

$$r_{(i)'}a_{(i)} = (\beta_i - \hat{\beta}_i)'X_i'a_{(i)} = (\beta_i - \hat{\beta}_i)' \sum_{s(i)} a_j x_{j-1}$$

$i = 1, 2$.

Lemma 2 follows from the corollary if we show

$$\sum_{s(i)} a_j x_{j-1} = O_p(T_i^{1/2}). \quad (8.2)$$

TABLE 3—Estimated autoregressive parameters.

		1	2	3	4	5	6	7	8	9
Order	Regime									
2	1	1.31	-.61							
2	2	1.37	-.67							
9	1	1.08	-.29	-.14	.06	-.06	-.01	.03	-.04	.20
9	2	1.24	-.49	-.12	.13	-.12	.10	-.09	.09	.10

Equation (8.2) follows from (8.3),

$$E \left(\sum_{s(i)} a_j x_{j-1} \right)' \left(\sum_{s(i)} a_k x_{k-1} \right) < \infty, \quad (8.3)$$

which we now establish.

Since a_j and x_{j-1} are independent and $Ea_j = 0$, one has

$$E(a_j a_k x_{j-1}' x_{k-1}) = \delta_{jk} E a_j^2 E(x_{j-1}' x_{j-1})$$

so that

$$E \left(\sum_{s(i)} a_j x_{j-1} \right)' \left(\sum_{s(i)} a_k x_{k-1} \right) = \sigma_i^2 \sum_{s(i)} E x_{j-1}' x_{j-1}.$$

Since the process is stationary for $t \leq T_1$ one has

$$E x_{j-1}' x_{j-1} = \text{Tr } \Sigma_1 < \infty$$

since Σ_1 is positive definite. This establishes (8.3) in the case $i = 1$. For $t \geq T_1$ the quantities $E x_t' x_t$ are not constant, but as in Anderson ([2], section 5.5, equation 12),

$$E(\tilde{x}_t - \mu_2 1_p)'(\tilde{x}_t - \mu_2 1_p)$$

$$\leq C_0 + C_1 E(\tilde{x}_{T_1} - \mu_2 1_p)'(\tilde{x}_{T_1} - \mu_2 1_p) \quad t \geq T_1$$

where $\tilde{x}_t$ is the first p components of x_t . Thus, $E(x_t' x_t)$ is bounded for $t > T_1$ which establishes (8.3) in the case $i = 2$, completing the proof.

Lemma 3:

$$\sqrt{\frac{T_1 T_2}{T_1 + T_2}} (F - \Delta^2) / \sqrt{2\Delta^2} \xrightarrow{L} N(0, 1 + \gamma_2/2).$$

Proof: The difference between the residual and error sum of squares is given by

$$\sum_{s(i)} (a_i^2 - \hat{a}_i^2) = \sum_{s(i)} (2r_i a_i + r_i^2) \quad i = 1, 2. \quad (8.4)$$

Lemma 2 shows that the right-hand side of (8.4) is bounded in probability for $i = 1, 2$. From the central limit theorem

$$T_2^{1/2} \left(\sum_{s(1)} (a_i^2 - \sigma_1^2) / T_1, \sum_{s(2)} (a_i^2 - \sigma_2^2) / T_2 \right) \xrightarrow{L} N(0, \Gamma)$$

where $\Gamma = (2 + \gamma_2) \text{diag}(\lambda \sigma_1^4, \sigma_2^4)$ and $\lambda = \lim T_2 / T_1$. The conclusion follows using the boundedness of (8.4) and Slutsky's Theorem.

Lemma 4: The Pitman a.r.e. of the Box-Andersen test with respect to the F is 1.

Proof: From (8.4) it follows that

$$\sum_{s(i)} \hat{a}_i^2 = \sum_{s(i)} a_i^2 (1 + O_p(T_i^{-1})) \quad i = 1, 2 \quad (8.5)$$

and as in Davis ([11], Section 2),

$$\sum_{s(i)} \hat{a}_i^4 = \sum_{s(i)} a_i^4 (1 + O_p(T_i^{-1/2})) \quad i = 1, 2. \quad (8.6)$$

TABLE 4—Test statistics and 2-sided p -values (in parentheses).

Statistic	Autoregressive order	
	2	9
F	1.64 (.04)	2.01 (<.01)
Box-Andersen	1.64 (.26)	2.01 (.07)
jackknife (m=1)	1.38 (.17)	1.64 (.10)

Using (8.5), (8.6) and the law of large numbers

$$d \xrightarrow{P} (1 + \gamma_2/2)^{-1}. \quad (8.7)$$

Using (8.7) the remainder of the proof is as in Shorack ([29], Section 5).

Proof of Theorem 1(i):

One has that

$$T_j^{1/2}(\hat{\theta}_j - g(\sigma_j^2)) = T_j^{1/2}(g(\hat{\sigma}_j^2) - g(\sigma_j^2)) - (T_j - 1) \sum_{s(j)} (g(v_{t(j)}) - g(\hat{\sigma}_j^2)) / T_j^{1/2} \quad (8.8)$$

Using the same argument as Davis ([11], Lemma 1), one can show that the second term on the right hand side of (8.8) converges in probability to 0. Using the boundedness of (8.4), (8.8), ([27], 6a.2 (ii)), and the multivariate form of Slutsky's Theorem, one has

$$T_2^{1/2}(\hat{\theta}_{(1)} - \log \sigma_1^2, \hat{\theta}_{(2)} - \log \sigma_2^2) \xrightarrow{L} N(0, (2 + \gamma_2)\Gamma_1)$$

where $\Gamma_1 = \text{diag}(\lambda, 1)$.

From Rao ([27], 6a.2(ii)),

$$T_2^{1/2}(\hat{\theta}_{(2)} - \hat{\theta}_{(1)} - \log \Delta^2) \xrightarrow{L} N(0, (2 + \gamma_2)(\lambda + 1)).$$

Using the same method as Davis ([11], Lemma 2), one has for differentiable g that

$$w_i / T_i \xrightarrow{P} g'(\sigma_i^2)^2 \sigma_i^4 (2 + \gamma_2)$$

so that with $g(u) = \log u$

$$w_i / T_i \xrightarrow{P} (2 + \gamma_2).$$

Thus,

$$T_2 \left(\sum_{i=1}^2 w_i / T_i (T_i - 1) \right) \xrightarrow{P} (\lambda + 1)(2 + \gamma_2)$$

so that an application of Slutsky's Theorem completes the proof.

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